

Extended Game Theory Notes

from Mathematical Optimization and Economic Theory

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1 Vertex Enumeration Fundamentals

1.1 Why a best response occurs at a vertex

A vertex of a polytope is a point that cannot be written as a nontrivial convex combination of other feasible points. By Carathéodory's theorem, any point in the convex hull of a set in $\mathbb{R}^d$ can be written as a convex combination of at most $d + 1$ vertices.

Claim

If p^{1*} is an equilibrium strategy that is not a vertex of Player 1's strategy polytope, then there exist vertices which are also equilibrium strategies. Furthermore, p^{1*} can be represented as a convex hull of these vertices.

Proof

Assume (p^{1*}, p^{2*}) constitute a Nash equilibrium, with p^{1*} an interior point of Player 1's strategy polytope. By Carathéodory's theorem, there exist a set of vertices v_i and positive weights w_i such that

$$p^{1*} = \sum_i w_i v_i, \quad \sum_i w_i = 1$$

Assume by contradiction that $v_k A(p^{2*})^\top < \max_{p^1 \in \mathbb{R}^m} p^1 A(p^{2*})^\top = p^{1*} A(p^{2*})^\top$ for some k . Then

$$p^{1*} A(p^{2*})^\top = \left(\sum_i w_i v_i \right) A(p^{2*})^\top = \sum_i w_i (v_i A(p^{2*})^\top)$$

$$\sum_i w_i (v_i A(p^{2*})^\top) = \sum_{i \neq k} w_i (v_i A(p^{2*})^\top) + w_k v_k A(p^{2*})^\top \leq \sum_{i \neq k} w_i (p^{1*} A(p^{2*})^\top) + w_k v_k A(p^{2*})^\top$$

$$\sum_{i \neq k} w_i (p^{1*} A(p^{2*})^\top) + w_k v_k A(p^{2*})^\top < \sum_{i \neq k} w_i (p^{1*} A(p^{2*})^\top) + w_k (p^{1*} A(p^{2*})^\top)$$

Since $\sum_i w_i (p^{1*} A(p^{2*})^\top) = p^{1*} A(p^{2*})^\top$, this is a contradiction.

The proof that p^{2*} is a vertex of Player 2's strategy polytope (or a convex hull of vertices that are equilibrium strategies) is analogous.

2 Best-Response Polytopes and Fully Labeled Vertices

Note that p^1 and p^2 are treated as column vectors rather than row vectors in this section.

2.1 Best-response polytopes

For a two player game with payoff matrices Π^1 and $\Pi^2 \in \mathbb{R}^{m \times n}$, define

$$P := \{p^1 \in \mathbb{R}^m : p^1 \geq 0, (\Pi^2)^T p^1 \leq \mathbf{1}\}, \quad Q := \{p^2 \in \mathbb{R}^n : p^2 \geq 0, (\Pi^1) p^2 \leq \mathbf{1}\}.$$

These are called the **best-response polytopes**. They encode the complementary slackness structure of Nash equilibrium.

Remark The construction of these polytopes assume that the maximum expected value for either player is 1. This is not necessarily the case, and reflects a scaling that is passed on to the probability vectors. The actual probability vectors can be recovered by dividing by the solution vectors by sum of their elements.

2.2 Labeling of vertices

Let $p^1 \in P$. The labels of p^1 are:

- $i \in \{1, \dots, n\}$ if $(p^1)_i = 0$,
- $j \in \{n+1, \dots, n+m\}$ if $((\Pi^2)^T p^1)_{j-n} = 1$.

Let $p^2 \in Q$. The labels of p^2 are:

- $i \in \{1, \dots, n\}$ if $(\Pi^1 p^2)_i = 1$.
- $j \in \{n+1, \dots, n+m\}$ if $(p^2)_{j-n} = 0$,

Thus there are $m+n$ total labels. A pair $(x, y) \in P_2 \times P_1$ is **fully labeled** if together they contain all $m+n$ labels. (p^1, p^2) is a Nash equilibrium (after normalization) if and only if (p^1, p^2) is a fully labeled pair of vertices.

The intuition behind a fully labeled pair of vertices is that, for each strategy of each player, that strategy either has the maximum expected value for that player or is played with probability 0. This is analagous to the Strong Minimax Theorem in zero-sum games.

3 Combinatorial Vertex Enumeration

3.1 Vertices of a polytope

Suppose a polytope in $\mathbb{R}^d$ is defined by M linear inequalities. A nondegenerate vertex of this polytope is formed by the intersection of exactly d linearly independent tight constraints.

In a strategy polytope, these potential vertices are obtained by:

1. Choosing d constraints,
2. Turning them into equalities,
3. Solving the square linear system,
4. Checking that the solution satisfies the other constraints

Number of candidates:

$$\binom{M}{d}.$$

3.2 Application to best-response polytopes

For Player 1, their strategy has m non-negativity constraints and n expected payoff constraints (for Player 2's possible strategies). Thus the maximum number of vertices in Player 1's strategy polytope is

$$\binom{m+n}{n}$$

Likewise, for Player 2's strategy polytope, the maximum number of vertices is

$$\binom{m+n}{m}$$

3.3 Support Enumeration vs Vertex Enumeration

- Support enumeration searches over candidate supports.
- Vertex enumeration searches over vertices of best-response polytopes.
- Both select components for a square linear system combinatorially and solve these systems
- The enumeration over vertices is probably more computationally intensive (I will code up vertex enumeration to check)

4 Lemke–Howson Algorithm

Algorithm. (Lemke–Howson)

Step 0. Initialize at origin vertices:

$$p^1 = \mathbf{0}, \quad p^2 = \mathbf{0}.$$

This pair is fully labeled since all nonnegativity constraints are tight.

Step 1. Choose an initial label k_0 and set $k = k_0$.

Step 2. Repeat:

1. Drop label k in the current polytope.
2. Pivot to the adjacent vertex.
3. Let k' be the duplicate label created. If $k' = k_0$, stop.
4. Set $k = k'$ and switch polytopes.

Step 3. Normalize (p^1, p^2) to obtain mixed strategies.

Remark. Lemke–Howson is a complementary pivot algorithm, analogous to simplex but operating across two polytopes.

5 Brown–von Neumann–Nash Dynamics

Recall Nash’s continuous functions:

$$\phi_{ij}(s) = \max(0, u_i(j, s_{-i}) - u_i(s)).$$

Note that s_{-i} refers to the strategies played by every player except for i .

The intuition is that these functions shift probabilistic mass to the pure strategies that outperform the current strategy.

Definition. The Brown–von Neumann–Nash (BNN) dynamics are given by:

$$\dot{s}_{ij} = \phi_{ij}(s) - s_{ij} \sum_k \phi_{ik}(s).$$

As previously discussed, every Nash equilibrium is a stationary point of these updating functions. Convergence holds in special classes (e.g. two-player zero-sum), but not necessarily in general games.